Yang Ouyang

Computer Science Ph.D. Student

youyang7@ncsu.edu (984) 325-2686 oyy2000.github.io Raleigh, NC 27603, United States

yang-ouyang-363425213 oyy2000

Professional Summary

Ph.D. student at North Carolina State University focusing on Efficient Inference-time LLM Reasoning and Trustworthy AI.

Education

North Carolina State University, Computer Science — Raleigh, U.S.A
Doctor of Philosophy — Aug 2024 – present
- Advisor: Jung-Eun Kim
- Research Interests: Efficient Inference-time LLM Reasoning and Trustworthy AI
- Award: 2026-2027 Provost's Doctoral Fellowship

Duke University, Electrical and Computer Engineering — Durham, U.S.A
Master of Engineering — Aug 2022 – May 2024
- GPA: 3.83 / 4.0
- Research Experience: Collaborated with Duke CEI on research projects in trustworthy AI.
- Award: Duke Kunshan University Scholarship Award (162K RMB)

Shenzhen University, Computer Science and Technology — Shenzhen, China
Bachelor of Engineering — Sept 2018 – July 2022
- GPA: 3.75 / 4.0
- Awards/Honor: Two-time winner of The Second Award of Studying Star in 2020 and 2021 (ranked 4 and 6); Outstanding Graduate of the Year 2022

Selected Publications

StepBack: Step-Level Error-Localized Resampling for Efficient Test-Time Reasoning 2026
Yang Ouyang, Jung-Eun Kim
(In Proceedings of NeurIPS 2026)

DenseSteer: Steering Small Language Models towards Dense Math Reasoning 2026
Yang Ouyang, Shuhang Lin, Jung-Eun Kim
(In Proceedings of ICML 2026)

Layer-Level Self-Exposure and Patch: Affirmative Token Mitigation for Jailbreak Attack Defense 2025
Yang Ouyang, Hengrui Gu, Shuhang Lin, Wenyue Hua, Jie Peng, Bhavya Kailkhura, Meijun Gao, Tianlong Chen, Kaixiong Zhou
(In Proceedings of NAACL 2025)

Min-K%++: Improved Baseline for Detecting Pre-Training Data of LLMs 2025
Jingyang Zhang, Jingwei Sun, Eric Yeats, Yang Ouyang, Martin Kuo, Jianyi Zhang, Hao Frank Yang, Hai Li
(In Proceedings of ICLR 2025)

Research Experience

In Proceedings of NeurIPS 2026, StepBack: Step-Level Error-Localized Resampling for Efficient LLM Reasoning — Raleigh, U.S.A — Jan 2026 – May 2026

- Proposed StepBack, a training-free inference-time decoding method that improves self-consistency by localizing likely first-error positions in chain-of-thought trajectories and resampling only uncertain suffixes instead of regenerating full reasoning paths.
- Designed lightweight step-level rollback signals using NLL increases and process reward model score drops to allocate test-time compute more efficiently within reasoning trajectories.
- Evaluated across math reasoning and multi-hop QA benchmarks; the PRM-based variant improved accuracy on 10 / 14 model-dataset pairs while reducing tokens in all settings, with up to 39.8% token savings over SC@32.

Project, Large-Scale Elastic Continued Pretraining for MoE Models — Shanghai, China — July 2026 – Sept 2026

- Adapted elastic training to a closed-source 15B-A2B MoE model using 1,024 GPUs, enabling three nested, weight-sharing subnetworks (3B, 9B, 15B) for different resource budgets.
- Explored five architectural dimensions (MoE depth, expert width, expert count, attention heads, and top-k routing) through budget-constrained sampling and importance-guided pruning, constructing nested subnetworks under joint total-parameter and active-parameter constraints.
- Optimized shared-weight continued pretraining across 11 benchmarks, outperforming size- and training-budget-matched standalone models by 1.6 and 3.9 points for small and large subnetworks, respectively.

In Proceedings of ICML 2026, DenseSteer: Steering Small Language Models towards Dense Math Reasoning — Raleigh, U.S.A — Aug 2025 – Jan 2026

- Proposed Dense Reasoning as a structural property of strong mathematical reasoning, empirically showing that higher reasoning density (fewer steps, more tokens per step) correlates with higher accuracy on GSM8K and OOD benchmarks.
- Developed a training-free, inference-time activation steering framework using only 50 rewrite-based contrastive pairs, avoiding distribution shift and preserving low token-level NLL.
- Achieved up to +5% GSM8K, +17.5% AMC, and +10% AIME accuracy gains on Qwen-2.5-3B-Instruct and LLaMA-3.2-3B-Instruct compared with heavy training-based distillation methods.

Internship Experience

Amazon Web Services | AI Infrastructure, Software Engineer Intern, DeepJavaLibrary Department — Mountain View (Remote), U.S.A — July 2022 – Oct 2022

- Developed core integration components between DJL-Serving and KServe, implementing 3 Java-based APIs for health monitoring, model introspection, and inference data processing within the model serving stack.
- Supported cloud-native model serving on Kubernetes by configuring deployment manifests, service endpoints, and runtime parameters for containerized inference workloads.

Technical Skills

Programming Languages: Python, Java, C++, JavaScript

Deep Learning Frameworks: PyTorch, Huggingface, Large-Scale Continued Pretraining for MoE LLM

AI-assisted Development Tools: Cursor, Codex, Claude Code